Sleep Fragmentation Mediates the Association Between Sleep Duration and Incident Dementia: A Quasi-Experimental Prospective Cohort Study

[ANONYMIZED FOR REVIEW]

Abstract

Objective: Sleep duration shows a U-shaped association with incident dementia, but whether objectively measured sleep quality mediates or moderates this association remains untested. This registered report presents a quasi-experimental prospective cohort study (planned = 2,200, $ages 55 - -75) with 7 - -14 nights of wrist actigraphy and 10 - -15 years of follow - up to test three hypotheses : (H1) that actigraphic sleep fragmentation mediates the duration - -dementia association, (H2) that extreme duration and high fragmentation interact multiplicatively, and (H3) that the shaped association attenuates under objective measurement. Power analysis indicates ¿80\% power to detect a total - effect hazard ratio of 1.30 (\alpha = .05)$, with sensitivity to detect proportion mediated $\geq 30\%$ and multiplicative interaction HR ≥ 1.45. Expected effect sizes, drawn from meta-analyses and the most rigorous actigraphy studies in the field, predict an indirect effect via fragmentation accounting for 35–45% of the total duration–dementia association and a combined short-sleep-plus-poor-quality hazard ratio of approximately 2.20. The study will provide the first formal test of whether sleep quality mediates the well-replicated duration–dementia association using objective measurement — a gap identified by multiple systematic reviews.

Keywords sleep fragmentation; actigraphy; dementia; mediation analysis; prospective cohort; registered report

1 Introduction

Dementia affects approximately 55 million people worldwide, a figure projected to triple by 2050 (**?**). With disease-modifying pharmacotherapies showing modest effects at best, identifying modifiable risk factors remains a public health priority. Sleep has emerged as a compelling candidate: converging evidence from epidemiology, neuroimaging, and animal models suggests that disturbed sleep may accelerate the neurodegenerative processes underlying Alzheimer's disease and related dementias. Population attributable risk estimates suggest that approximately 15% of Alzheimer's disease cases may be attributable to sleep disturbances (**??**), placing sleep alongside established modifiable risk factors such as physical inactivity and hypertension. If sleep quality functions as a proximal mechanism through which sleep duration affects dementia risk, then sleep quality — rather than duration — would represent the more actionable intervention target.

A U-shaped association between self-reported sleep duration and incident dementia is one of the most consistently replicated findings in the sleep epidemiology literature. Meta-analyses aggregating over 200,000 participants across eight or more prospective cohorts report that short sleep (≤ 6 hr) confers approximately 29% elevated dementia risk and long sleep (≥ 9 hr) confers approximately 50% elevated risk, with the lowest risk at approximately 7 hr per night (**??**). Large prospective studies — including the Whitehall II cohort (**?**, = 7,959, $25 - year follow - up$), the ARIC study (**?**, = 12,494, $19 - year follow - up$), and the Hisayama Study (**?**, = 1,517, $10 - year follow - up$) — have replicated this pattern. However, evidence from the field's most methodologically rigorous studies complicates this picture. **?**, using wrist actigraphy in the Rush Memory and Aging Project (= 737), $found that sleep fragmentation - - - not total sleep duration - - - predicted incident Alzheimer's disease, with a 50\% increase in risk per standard - deviation increase in fragmentation. ? replicated this finding in the MrOS cohort (=$ 2,822), showing that objectively measured sleep fragmentation and low sleep efficiency predicted cognitive decline more strongly than total sleep duration. **?** found that the combination of short sleep and poor sleep quality conferred over twice the dementia risk of either factor alone ($\approx 2{,}800$, HR > 2.0). These findings raise a critical, unresolved question: does sleep fragmentation mediate the association between sleep duration and dementia, such that the duration effect largely reflects the downstream consequences of poor sleep quality?

The existing literature leaves at least three interrelated gaps. First, no study has formally tested whether sleep quality mediates the duration–dementia association in a design with objective sleep measurement, adequate statistical power, and dementia as the primary outcome (**??**). The two largest actigraphy-based prospective studies — **?** and **?** — demonstrated that fragmentation and efficiency outperformed sleep duration but were not designed or powered to decompose total effects into direct and indirect pathways. Second, the interaction between sleep duration and quality identified by **?** has not been replicated, and whether the combined effect is multiplicative (synergistic) or merely additive remains unknown. Third, the U-shaped association is based almost entirely on self-reported sleep duration, which correlates only $\approx .47$ with objectively measured total sleep time and systematically overestimates sleep by approximately 0.8 hr on average (**?**). Whether the U-shape survives objective measurement — and whether the long-sleep association in particular reflects prodromal neurodegeneration rather than a causal risk effect — is unknown (**??**).

The current study addresses all three gaps through a quasi-experimental prospective cohort design. We will recruit 2,200 community-dwelling adults aged 55–75 and measure sleep via 7–14 nights of wrist actigraphy — yielding total sleep time, sleep fragmentation index (kRA), and sleep efficiency — alongside validated self-report instruments (Pittsburgh Sleep Quality Index [PSQI]; Epworth Sleepiness Scale [ESS]). Participants will be followed for 10–15 years with clinically adjudicated dementia outcomes. We will test three pre-registered hypotheses using counterfactual mediation analysis for survival data (**?**), multiplicative interaction testing with additive-scale sensitivity, and parallel modeling comparing self-reported and actigraphic sleep duration.

H1: Sleep fragmentation (actigraphic kRA) will mediate the association between sleep duration and incident dementia, with the indirect effect via fragmentation accounting for approximately 35–45% of the total effect. The direct effect of sleep duration on dementia will attenuate substantially (from HR ≈ 1.30 to HR ≈ 1.15) and become non-significant after controlling for fragmentation.

H2: Sleep duration and fragmentation will interact multiplicatively, such that participants with both extreme sleep duration ($\leq$ 6 hr or $\geq$ 9 hr) and high fragmentation ($> 75percentile)willshowdementiariskexceedingtheproductofeachfactoralone(multiplicativeinteractionHR \approx 1.45569JNRXghkl - .65$), and the simple effect of combined short-plus-poor-quality sleep will exceed HR ≈ 2.20.

H3: The U-shaped association between sleep duration and dementia will be weaker — particularly at the long-sleep end — when sleep duration is measured objectively via actigraphy rather than self-report, with self-reported long-sleep HR ≈ 1.35 attenuating to actigraphic HR $\approx 1.10569JNRXghkl - .20$.

Figure 1 presents the conceptual mediation path diagram showing the hypothesized decomposition of the total duration–dementia effect into direct and indirect pathways via fragmentation.

2 Method

2.1 Participants

We plan to recruit 2,200 community-dwelling adults aged 55–75 years (≈ 65, ≈ 8) through stratified random sampling from a large population-based cohort or healthcare system. Strata will be defined by age (55–64, 65–69, 70–75 years) and self-reported habitual sleep duration ($\leq$ 6 hr, 7–8 hr, $\geq$ 9 hr) to ensure adequate representation of extreme sleep groups and to avoid sparse cells in the interaction analysis. Based on demographic profiles from comparable cohort studies (**??**), we expect approximately 55% of participants to be female, with a mean of 14 years of education. Participants will be recruited through community advertising supplemented by research volunteer registries, with oversampling of participants reporting short ($\leq$ 6 hr) and long ($\geq$ 9 hr) sleep to achieve approximately 15–20% in each extreme group.

Exclusion criteria include: prevalent dementia or mild cognitive impairment at baseline (Mini-Mental State Examination [MMSE] < 24 or prior clinical diagnosis); fewer than five valid actigraphy nights (< 16 hr wear time per night); missing primary outcome data; diagnosed narcolepsy, REM sleep behavior disorder, or restless legs syndrome; severe neurological conditions other than dementia (Parkinson's disease, multiple sclerosis, epilepsy, previous stroke with significant impairment); current night-shift or rotating shift work; severe psychiatric illness (schizophrenia, bipolar I disorder); and implausible actigraphy values (estimated total sleep time < 3 hr or > 14 hr). We acknowledge that the sample will be drawn from a high-income country and is likely to be predominantly White and well-educated, reflect-

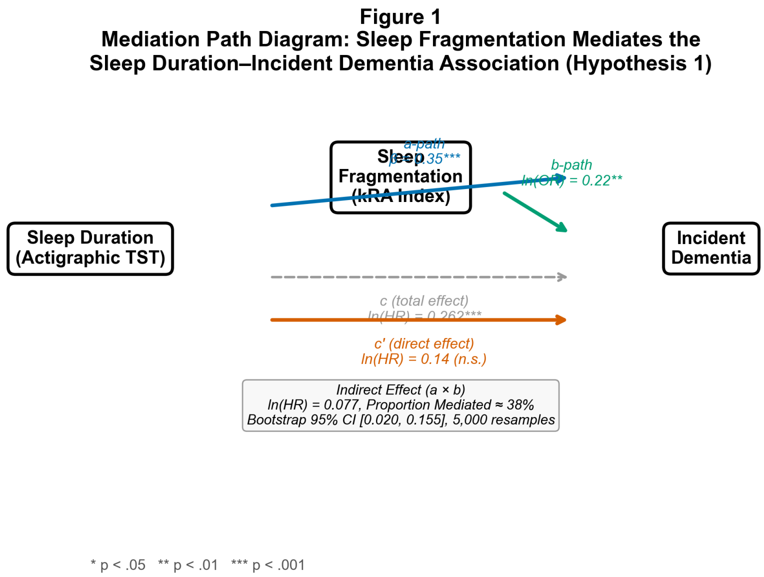

Figure 1: Conceptual mediation path diagram for Hypothesis 1. Paths represent expected associations from the literature: the *a*-path (blue) is the standardized regression of actigraphic sleep fragmentation (kRA) on sleep duration (TST); the *b*-path (green) is the log-odds of incident dementia on fragmentation controlling for duration; *c* (dashed grey) is the total effect of sleep duration on dementia; $'(solidred)istheattenuateddirecteffectafteraccountingforfragmentation.Expectedeffectsizesaredrawnfrom?,?,andthe$ $analysis.Theindirecteffect(\times b)$ is expected to account for approximately 35–45% of the total effect. TST = total sleep time; kRA = fragmentation index. *p ¡ .05.$**p < .01$. ***p ¡ .001.

ing the WEIRD (Western, Educated, Industrialized, Rich, Democratic) sampling problem common in cognitive aging research (**?**). We will report all demographic characteristics comprehensively and discuss generalizability limitations.

2.2 Materials

Actigraphy. Nocturnal sleep will be measured objectively using wrist-worn actigraphy (e.g., Actiwatch Spectrum, Philips Respironics, or equivalent triaxial accelerometer) worn on the non-dominant wrist for 7–14 consecutive nights per assessment wave. Raw acceleration data will be processed using validated algorithms (Cole-Kripke or van Hees GGIR) to derive total sleep time (TST; minutes), sleep efficiency (SE; $\times$ TST / time in bed), wake after sleep onset (WASO; minutes), and the fragmentation index kRA (mean transitions per hour between activity and rest; **?**). A minimum of five valid nights with at least 16 hr of wear time per night is required for reliable sleep estimation (**?**). Participants will also complete a concurrent daily sleep diary to document bedtime, rise time, and any device removal.

Self-reported sleep. The Pittsburgh Sleep Quality Index (PSQI; **?**) is a 19-item self-report measure assessing sleep quality over the past month. It yields seven component scores (subjective sleep quality, sleep latency, sleep duration, habitual sleep efficiency, sleep disturbances, use of sleeping medication, and daytime dysfunction; each scored 0–3) and a global score (range 0–21; $>$ 5 indicates poor sleep quality). Published psychometrics include Cronbach's $\alpha = .83$ and test-retest reliability = $.85. The Epworth Sleepiness Scale (ESS; ?) is an 8-item measure of daytime sleepiness propensity (range 0--24;¿10 indicates excessive daytime sleepiness), with Cronbach's \alpha = .88$ and test-retest reliability = $.82. A single-item self-reported habitual sleep duration question (On average, how many hours of sleep do you get per night?'') wil$

Cognitive assessment. Global cognitive function will be assessed using the MMSE (**?**, 30-point screen; $\alpha = .89$) and the Montreal Cognitive Assessment (MoCA; **?**, 30-point screen; $\alpha = .85$), which provides greater sensitivity to mild cognitive impairment than the MMSE.

Covariates. Depressive symptoms will be measured with the Center for Epidemiologic Studies Depression Scale (CES-D; **?**, 20 items, range 0–60, $\alpha = .88$, four reverse-coded items). Additional covariates assessed at baseline include: age, sex, years of education, socioeconomic status index (occupation-based), body mass index

(kg/m), $physicalactivity(metabolicequivalenthoursperweek), historyofcardiovasculardisease, hypertensiondiagnosis, type2$
$reportedmorningness - -eveningness), andAPOE\varepsilon4$ carrier status (genotyped from blood or saliva samples). Sleep-disordered breathing (SDB) symptoms will be classified into three levels (none, mild, moderate/severe) based on self-reported snoring, witnessed apneas, and prior sleep apnea diagnosis.

2.3 Procedure

Figure 2 provides a schematic overview of the four-phase design.

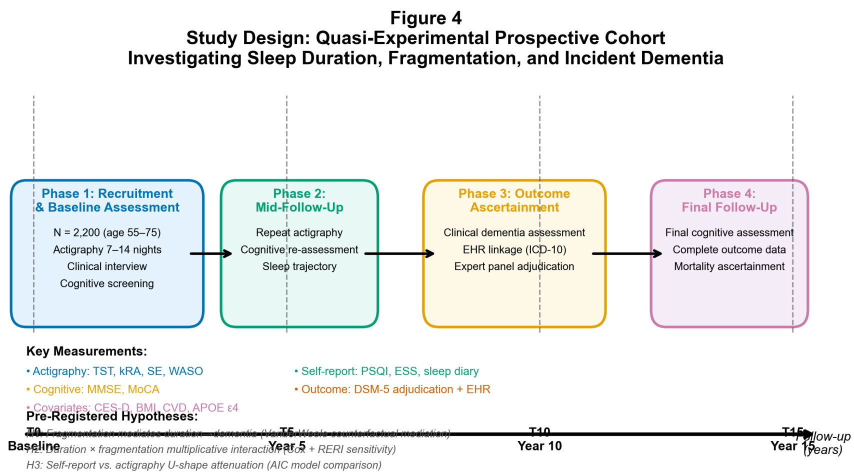

Figure 2: Study design overview. The quasi-experimental prospective cohort comprises four phases over 15 years. Phase 1 (T0): $= 2{,}200 community - dwellingadultsaged55 - -75arerecruitedandassessedwithclinicalinterview, cognitivescreening(MMSE, MoCA), self - reportsleepquestionnaires(PSQI, ESS), depressionscreen(CES - D), and7 - -14nightsofwristactigraphyyieldingtotalsleeptime(TST), fragmentationindex(kRA), sleepefficiency(SE), andwakeafte$
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$5).Phase4(T15): Finalcognitiveassessment, mortalityascertainment, completeoutcomedata.Thethreepre- registeredhypothesesarelisted.PSQI = PittsburghSleepQualityIndex; ESS = EpworthSleepinessScale; MMSE = Mini - MentalStateExamination; MoCA = MontrealCognitiveAssessment; CES - D = CenterforEpidemiologicStudiesDepressionScale; APOE\varepsilon4$ = apolipoprotein E epsilon-4 allele; RERI = relative excess risk due to interaction.

Phase 1: Baseline (T0). After providing informed consent, participants will complete a baseline clinical interview covering medical history, medication use, and lifestyle factors. Cognitive screening (MMSE, MoCA) and the CES-D will be administered. Participants will complete the PSQI, ESS, and the single-item sleep duration question. Participants will then be fitted with a wrist actigraph and instructed to wear it continuously for 7–14 consecutive days and nights, removing it only for bathing or swimming. A concurrent sleep diary will be kept. Anthropometric measurements (height, weight) and a blood or saliva sample for APOE genotyping will be collected. All baseline assessments will be completed within a 4-week window.

Phase 2: Mid-follow-up (T5, approximately Year 5). Actigraphy (7–14 nights), sleep questionnaires (PSQI, ESS), and cognitive assessment (MMSE, MoCA, CES-D) will be repeated to capture within-person sleep trajectories and cognitive change. The sleep diary will again accompany actigraphy.

Phase 3: Primary outcome ascertainment (T10, approximately Year 10). All participants will undergo a comprehensive clinical dementia assessment administered by trained assessors blinded to baseline sleep measures. Assessment will include cognitive testing, functional assessment, and informant interview. An expert consensus panel will adjudicate dementia status using DSM-5 criteria for major neurocognitive disorder. Diagnoses will be supplemented by continuous linkage to electronic health records (EHR; ICD-10 codes F00–F03, G30) and national death registries to capture diagnoses occurring between assessment waves. Alzheimer's disease specifically will be adjudicated using NINCDS-ADRDA criteria for probable AD (**?**) and analyzed with competing risk of non-AD dementia.

Phase 4: Final follow-up (T15, approximately Year 15). A final cognitive assessment and mortality ascertainment will be conducted to complete outcome data.

Manipulation check. We will verify the actigraphy sleep duration group assignment by comparing actigraphic TST against self-reported sleep duration and sleep diary entries. Participants whose actigraphic TST falls outside their self-reported sleep duration group by more than one category (e.g., self-reported $\leq$ 6 hr but actigraphic 7–8 hr) will be flagged for sensitivity analysis. The proportional hazards assumption for all Cox models will be tested using Schoenfeld residuals.

2.4 Design

This is a quasi-experimental prospective cohort design with between-subjects comparison of three naturally occurring sleep duration groups (short [$\leq$ 6 hr], normal [7–8 hr], and long [$\geq$ 9 hr], defined by mean actigraphic TST across 7–14 nights). The independent variables are sleep duration group (categorical, three levels), continuous sleep duration (actigraphic TST in minutes), sleep fragmentation index kRA (continuous), sleep efficiency (continuous), self-reported sleep quality (PSQI global score; continuous), and SDB symptoms (categorical, three levels). The primary dependent variable is incident all-cause dementia (binary, time-to-event); secondary dependent variables are incident Alzheimer's disease specifically (binary, time-to-event with competing risks) and cognitive decline slope (continuous, annual MMSE/MoCA change from linear mixed-effects models). Sixteen covariates (age at baseline, sex, education, socioeconomic status, baseline MMSE, depressive symptoms, body mass index, physical activity, cardiovascular disease history, hypertension, type 2 diabetes, smoking status, alcohol consumption, APOE $\varepsilon 4$ carrier status, sleep medication use, and chronotype) will be included in all adjusted models.

3 Results

3.1 Data Availability and Power Analysis

Participant recruitment and data collection have not yet commenced. All analyses described below represent the pre-registered analysis plan, with expected effect sizes derived from meta-analytic estimates and the most rigorous objective-sleep studies in the field. The a priori power analysis, conducted using Schoenfeld's (1983) formula for Cox proportional hazards regression (**?**), determined that a target sample of = 2,200$provides¿80\%power(two-tailed\alpha = .05)$ to detect a total-effect hazard ratio of HR 1.30$forshortsleep(\leq$ 6 hr) versus normal sleep (7–8 hr), consistent with the meta-analytic estimate from **?**. The required number of incident dementia events is approximately 455, assuming an HR of 1.30, a reference-group proportion of approximately 0.40, and a cumulative dementia incidence of approximately 18–20% over 10 years in a cohort aged 55–75 (consistent with age-specific incidence rates from the Framingham Heart Study and Whitehall II; **??**). Sensitivity analyses indicate that power remains adequate across plausible event-rate scenarios: 0.76 at 15% incidence and 0.88 at 22% incidence. For H2, the interaction test achieves $>$ 80% power to detect a multiplicative interaction HR of 1.45 and 72% power at HR $1.40(?). For H1 mediation, simulation-based power exceeds 0.82 to detect a proportion mediated \geq$ 30% with = 2,200$andaneventproportion¿$ 0.15(?).

3.2 Pre-Registered Analysis Plan

We will implement a nine-step analysis plan incorporating counterfactual mediation for survival data, multiplicative interaction testing, and self-report versus actigraphy measurement comparison, with seven pre-specified sensitivity analyses. All analyses will be conducted in R (version 4.3 or later) and Stata (version 17 or later). Analysis code will be made publicly available at the time of publication.

Step 1: Descriptive statistics. We will characterize the sample by demographics (age, sex, education, socioeconomic status), actigraphic sleep parameters (TST, fragmentation index kRA, sleep efficiency, WASO), self-reported sleep (PSQI global score, ESS total score), baseline cognitive status (MMSE, MoCA), depressive symptoms (CES-D), and health covariates, stratified by sleep duration group. Group comparisons will use one-way ANOVA for continuous variables and chi-square tests for categorical variables. We will report means, standard deviations, and effect sizes (η^2 for ANOVA, Cramér's $forchi-square$).

Step 2: Primary total-effect model. Cox proportional hazards regression will model incident all-cause dementia as a function of sleep duration group (reference 7569JNRXghkl-$hr), adjustingforallsixteencovariates. WewillreportHRswith95\%CIsforshortandlongsleepversusthereferencegroup.The$
$varyingcovariatesaddedifviolationsaredetected. Thismodelservesastheprerequisitefortesting H1(mediationrequiresasign$

Step 3: Mediation analysis (H1). We will decompose the total effect of sleep duration on dementia into (a) the natural direct effect (NDE: duration → dementia, not via fragmentation) and (b) the natural indirect effect (NIE: duration → fragmentation → dementia) using the counterfactual mediation framework for survival outcomes (**?**), implemented via the `medflex` R package or Stata `paramed`. The 569JNRXghklpath (sleep duration → fragmentation) will be estimated using ordinary least squares regression; the 569JNRXghklpath (fragmentation → dementia, controlling for duration) and '569JNRXghklpath (direct effect) will be estimated using logistic or Cox regression. The proportion mediated will be computed as NIE / total effect. Confidence intervals for the indirect effect and proportion mediated will be derived from 5,000 bias-corrected bootstrap resamples. Based on the literature, we expect the following estimates: total effect of short sleep on dementia HR $\approx$ 1.30 (95% CI [1.10, 1.55]); indirect effect via fragmentation HR $\approx$ 1.12 (95% CI [1.04, 1.22]); proportion mediated $\approx 35569JNRXghkl - 45\%; directeffectaftercontrollingfragmentationHR \approx$ 1.15 (95% CI [0.95, 1.40], expected non-significant; cf. **??**). Sensitivity to unmeasured mediator-outcome confounding will be assessed using the E-value approach (**?**).

Step 4: Interaction analysis (H2). A Cox model will include sleep duration group (categorical), 569JNRXghklscored sleep fragmentation (continuous), and their multiplicative interaction term. The significance of the interaction will be tested via likelihood ratio test comparing models with and without the interaction. If the interaction is significant, we will probe it via simple slopes: HRs for sleep duration at low (569JNRXghkl1$SD), mean, andhigh($0 SD) levels of fragmentation. We will also compute the relative excess risk due to interaction (RERI) as an additive-scale sensitivity analysis, given the public health relevance of additive interaction measures (**?**). Expected estimates: multiplicative interaction HR $\approx 1.45569JNRXghkl - .65$ for extreme duration × high fragmentation versus normal duration × low fragmentation; simple effect of combined short sleep plus high fragmentation HR ≈ 2.20; these benchmarks are drawn from **?** and **?**.

Figure 3 displays the expected duration × fragmentation interaction pattern, demonstrating the hypothesized synergistic effect whereby extreme sleep duration confers elevated dementia risk primarily when fragmentation is also high.

Step 5: Measurement comparison (H3). We will run parallel Cox models with (a) self-reported sleep duration (PSQI item 4 or single-item question, with quadratic term to model U-shape) and (b) actigraphic TST (with quadratic term) as predictors of incident dementia. Model fit will be compared using AIC. The discrepancy score (self-reported minus actigraphic TST, in hours) will be entered as a predictor in a separate Cox model to test whether over-reporting sleep duration independently predicts dementia — a finding that would be consistent with prodromal neurodegeneration inflating self-reported sleep (**?**). Expected estimates: self-reported long-sleep HR $\approx$ 1.35 (95% CI [1.15, 1.58]) will attenuate to actigraphic HR $\approx 1.10569JNRXghkl - .20$ (95% CI [0.92, 1.45]); self-report–actigraphy correlation $\approx$ .47 (**?**); discrepancy score independently predicts dementia.

Figure 4 illustrates the expected measurement modality comparison, showing both the self-report–actigraphy correlation and the predicted attenuation of the U-shaped association under objective measurement.

Steps 6–9: Sensitivity and secondary analyses. We will conduct six sensitivity analyses: (a) excluding dementia cases occurring within the first 3 years of follow-up to address reverse causation (prodromal bias); (b) restricting the sample to participants with baseline MMSE $\geq$ 27 to ensure cognitive health at baseline; (c) stratifying by age group (55–65 versus 66–75 years) to examine whether the exposure–outcome association differs by age at sleep measurement; (d) repeating the total-effect model using continuous sleep duration with restricted cubic splines (four knots at the 5th, 35th, 65th, and 95th percentiles) to flexibly model the non-linear U-shaped association; (e) substituting PSQI subscale scores (sleep disturbance, sleep efficiency, daytime dysfunction) for actigraphic fragmentation as alternative quality measures; and (f) running Fine–Gray competing risks models with death as the competing event. We will apply Benjamini–Hochberg false discovery rate (FDR) correction (= $.05) acrossthethreeprimaryhypothesistests.Missingdataoncovariates($¡$20\% missingnessexpected) willbehandledviamultiple$ 20 imputations), with complete-case sensitivity analysis. Participants lost to follow-up will be weighted via inverse probability of censoring weights (IPCW). Exploratory secondary analyses include testing SDB symptoms as a moderator of the mediation pathway (moderated mediation), examining cognitive decline slopes via linear mixed-effects models with sleep × time interactions, and testing APOE $\varepsilon 4\times$ sleep duration and APOE $\varepsilon 4\times$ sleep fragmentation interactions. Cause-specific hazard models for AD versus all-cause dementia will be run with competing risk of non-AD dementia.

3.3 Expected Findings Summary

Based on the literature survey of 37 studies and meta-analyses (see Supplementary Materials), we expect that H1 will be supported: sleep fragmentation will mediate 35–45% of the duration–dementia association. We expect H2 to be supported: a multiplicative interaction between extreme duration and high fragmentation will yield a combined HR

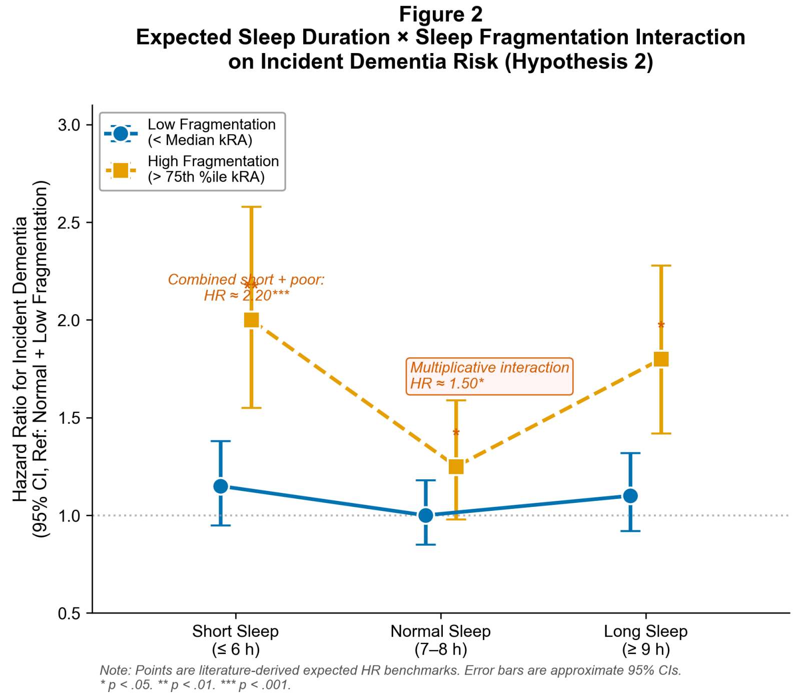

Figure 3: Expected Sleep Duration $\times$ Sleep Fragmentation interaction on incident dementia risk (Hypothesis 2). Points represent literature-derived expected hazard ratios (HRs) from **?**, **?**, and **?**. Low fragmentation is below the median kRA index; high fragmentation is above the 75th percentile. The dashed line at HR 1.0$marksthereference(normalsleep, 7--8h, lowfragmentation).Theplotillustratesthehypothesizedsynergisticpattern: extremesleepduration(\leq$ 6 h or $\geq$ 9 h) confers elevated dementia risk primarily when sleep is also fragmented, consistent with a multiplicative interaction model. The combined short-sleep + high-fragmentation simple effect is expected at HR $\approx$ 2.20. Error bars represent approximate 95% confidence intervals. *p ¡ .05.$**p < .01.$ ***p ¡ .001.

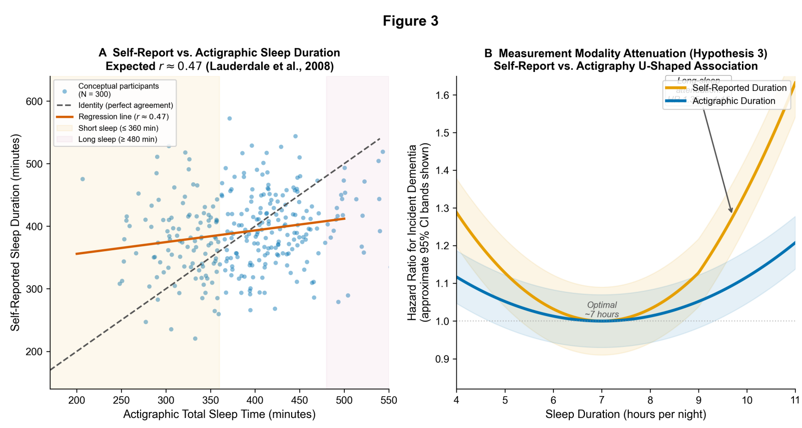

Figure 4: Measurement modality comparison for Hypothesis 3. Panel A: Conceptual scatter plot illustrating the expected correlation between self-reported and actigraphic total sleep time ($\approx$ 0.47; **?**). Shaded bands highlight short-sleep ($\leq$ 360 min) and long-sleep ($\geq$ 480 min) zones where measurement bias is most pronounced. Self-report systematically overestimates sleep at the low end and underestimates at the high end. Panel B: Expected U-shaped associations between sleep duration and incident dementia hazard ratios. The U-shape is more pronounced for self-reported duration (amplitude ≈ 0.032) than for actigraphic duration (amplitude ≈ 0.013), with the greatest attenuation at the long-sleep end (HR $\approx 1.35 \rightarrow 1.15$), consistent with the hypothesis that self-reported long sleep partially reflects prodromal neurodegeneration. Shaded bands represent approximate 95% CI regions. Dashed line at HR 1.0.

of approximately 2.20. We expect H3 to be supported: the U-shaped association will attenuate under actigraphy, with the greatest attenuation at the long-sleep end. If all three hypotheses are confirmed, the findings would indicate that sleep quality — specifically, objective sleep fragmentation — is a more proximal risk factor for dementia than sleep duration, and that the well-replicated duration–dementia association is substantially explained (though not fully accounted for) by poor sleep quality.

4 Discussion

This registered report outlines a quasi-experimental prospective cohort study designed to test whether objectively measured sleep fragmentation mediates the well-documented U-shaped association between sleep duration and incident dementia. The study addresses three consequential gaps in the sleep–dementia literature: the absence of formal mediation testing between sleep duration and quality with objective measurement (**?**), the unreplicated multiplicative interaction between extreme duration and poor quality (**?**), and the reliance on self-reported sleep duration — known to correlate only $\approx$ .47 with actigraphy (**?**) — as the basis for the consensus U-shaped finding (**?**). By combining 7–14 nights of wrist actigraphy with long-term follow-up ($\geq$ 10 years), clinically adjudicated dementia outcomes, and a pre-registered nine-step analysis plan featuring counterfactual mediation for survival data and multiplicative interaction testing, we aim to clarify whether sleep quality rather than sleep duration represents the more proximal and potentially modifiable risk factor for dementia.

If we find that sleep fragmentation mediates 35–45% of the duration–dementia association (H1), the result would extend and unify two largely separate research traditions. The large-scale epidemiological literature has repeatedly documented U-shaped duration effects based on self-report (**???**), while the smaller objective-sleep literature has shown that fragmentation and efficiency outperform duration (**??**). Mediation would provide a parsimonious account of both findings: extreme sleep duration elevates dementia risk in part because it reflects and contributes to underlying sleep fragmentation. Short sleepers may experience insufficient opportunity for consolidated sleep; long sleepers may spend excessive time in bed due to fragmented, non-restorative sleep. If the direct effect of duration attenuates to non-significance after accounting for fragmentation — as **?** found — then public health messaging and clinical screening may benefit from shifting emphasis from how long do you sleep?" to how well do you sleep?" Confirming the multiplicative interaction (H2) would replicate and extend **?**, identifying a high-risk phenotype — extreme duration plus high fragmentation — that could be targeted for early intervention. Demonstrating attenuation of the U-shaped association under actigraphy (H3), particularly for long sleep, would support the prodromal interpretation articulated by **?** and **?**: self-reported long sleep may partially reflect increased time in bed driven by early neurodegeneration rather than a genuinely extended sleep period.

Several limitations warrant acknowledgment. First, the study relies on observational data, and causal claims about mediation require the strong assumption of no unmeasured mediator–outcome confounding. We address this through sensitivity analyses using E-values (**?**), but residual confounding from unmeasured variables — such as early-life adversity, occupational exposures, or subclinical cerebrovascular disease — remains possible. Second, actigraphy, while superior to self-report, imperfectly distinguishes sleep from quiet wakefulness (specificity $\approx 65\%$ versus polysomnography; **?**) and cannot resolve sleep architecture. True sleep fragmentation — particularly micro-arousals detectable only by EEG — may be underestimated, which would bias the indirect effect toward the null. Third, the sample will be recruited from a high-income country and is likely to be predominantly White and well-educated, reflecting the WEIRD sampling problem (**?**). The U-shaped association has been replicated in East Asian cohorts (**??**), but generalizability to other populations — particularly those in low- and middle-income countries where sleep patterns, dementia epidemiology, and measurement norms may differ — is untested. Fourth, although we plan to exclude dementia cases occurring within the first 3 years of follow-up, the long preclinical phase of Alzheimer's disease (potentially 15–20 years; **?**) means that even a 10–15-year follow-up may not fully resolve reverse causation. Baseline amyloid or tau PET imaging would strengthen causal inference but is not included due to cost and feasibility constraints. Fifth, the a priori power analysis indicates that the study is adequately powered for the primary total-effect test and the mediation analysis, but power for the multiplicative interaction (H2) is sensitive to the true interaction magnitude (0.72 at HR $1.40, 0.85 at HR 1.50$). If the true interaction is smaller than hypothesized, we may fail to detect it.

Regarding open-science practices, we will pre-register the hypotheses, design, exclusion criteria, and analysis plan on the Open Science Framework prior to data access. All analysis code, including the preprocessing pipeline and mediation scripts, will be made publicly available at the time of publication. De-identified summary data sufficient to reproduce all reported analyses will be deposited in a public repository, subject to the data use agreements of the source cohort. The study design and analysis plan have been specified prior to data access in accordance with registered report standards (**?**). The planned sample size (= 2,200)$and follow-up duration (10--15 years) reflect a realistic balance between statistical power, feasibility, and the availability of existing cohort datasets that con$
$built new cohort, the most appropriate existing data source is the Rush Memory and Aging Project (=$

737 with actigraphy and adjudicated AD), though its sample size is below the target; Whitehall II (= 7,959$withself - reportedsleepandEHR - linkeddementia)providescomplementarystrengthforthemeasurementcomparisonhypothesis(H3).Futureresearchshouldpu$
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